```
mysql> create database libraryManagement;
Query OK, 1 row affected (0.01 sec)

mysql> show databases;
+--------------------------+
| Database                 |
+--------------------------+
| datamodels               |
| info                     |
| information_schema       |
| library_management_system |
| librarymanagement        |
| mysql                    |
| performance_schema       |
| university               |
| vish                     |
+--------------------------+
9 rows in set (0.00 sec)

mysql> use librarymanagement;
Database changed
mysql> create table Authors(author_id int primary key, name varchar(30), country varchar(30));
Query OK, 0 rows affected (0.05 sec)

mysql> create table Books(book_id int primary key, title varchar(50),author_id int,published_year year, available_copies int, foreign key (author_id) refere
nces Authors(author_id));
Query OK, 0 rows affected (0.13 sec)

mysql> create table Members(member_id int primary key, name varchar(100), email varchar(100) unique, phone varchar(15));
Query OK, 0 rows affected (0.12 sec)

mysql> create table Borrow_Records(record_id int primary key, member_id int,book_id int , borrow_date date, return_date date, foreign key (member_id) refere
nces Members(member_id), foreign key (book_id) references Books(book_id));
Query OK, 0 rows affected (0.05 sec)

mysql> Alter table books
    -> add genre varchar(50);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> desc Authors;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| author_id | int         | NO   | PRI | NULL    |       |
| name      | varchar(30) | YES  |     | NULL    |       |
| country   | varchar(30) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
3 rows in set (0.07 sec)

mysql> desc Books;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| book_id          | int         | NO   | PRI | NULL    |       |
| title            | varchar(50) | YES  |     | NULL    |       |
| author_id        | int         | YES  | MUL | NULL    |       |
| published_year   | year        | YES  |     | NULL    |       |
| available_copies | int         | YES  |     | NULL    |       |
| genre            | varchar(50) | YES  |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> desc Members;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| member_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| email     | varchar(100) | YES  | UNI | NULL    |       |
| phone     | varchar(15)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> desc Borrow_Records;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| record_id   | int  | NO   | PRI | NULL    |       |
| member_id   | int  | YES  | MUL | NULL    |       |
```

```
| book_id     | int  | YES  | MUL | NULL    |       |
| borrow_date | date | YES  |     | NULL    |       |
| return_date | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> insert into Authors(author_id, name, country) values (1, 'R. K. Narayan', 'India'),(2, 'J. K. Rowling', 'United Kingdom'),(3, 'George Orwell', 'Unite
d Kingdom');
Query OK, 3 rows affected (0.07 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from Authors;
+-----------+--------------+----------------+
| author_id | name         | country        |
+-----------+--------------+----------------+
|         1 | R. K. Narayan | India          |
|         2 | J. K. Rowling | United Kingdom |
|         3 | George Orwell | United Kingdom |
+-----------+--------------+----------------+
3 rows in set (0.00 sec)

mysql> insert into Books(book_id, title, author_id, published_year, available_copies, genre) values (1, 'Database Systems', 1, 2018, 5, 'Technology'),(2, 'H
arry Potter', 2, 1997, 8, 'Fantasy'),(3, 'Animal Farm', 3, 1945, 4, 'Political Fiction'),(4, '1984', 3, 2019, 6, 'Dystopian'),(5, 'SQL Fundamentals', 1, 202
2, 10, 'Technology');
Query OK, 5 rows affected (0.07 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Books;
+---------+-----------------+-----------+----------------+-----------------+------------------+
| book_id | title           | author_id | published_year | available_copies | genre            |
+---------+-----------------+-----------+----------------+-----------------+------------------+
|       1 | Database Systems |         1 |           2018 |               5 | Technology       |
|       2 | Harry Potter     |         2 |           1997 |               8 | Fantasy          |
|       3 | Animal Farm      |         3 |           1945 |               4 | Political Fiction |
|       4 | 1984             |         3 |           2019 |               6 | Dystopian        |
|       5 | SQL Fundamentals |         1 |           2022 |              10 | Technology       |
+---------+-----------------+-----------+----------------+-----------------+------------------+
5 rows in set (0.07 sec)

mysql> insert into Members(member_id, name , email, phone) values (1, 'vishva', 'vishva@gmail.com','9952364875'), (2, 'Parii', 'pari@gmail.com', '7845963254
'), (3,'Sakshi', 'chaudhari@gmail.com','425935845');
Query OK, 3 rows affected (0.07 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from Members;
+-----------+--------+--------------------+------------+
| member_id | name   | email              | phone      |
+-----------+--------+--------------------+------------+
|         1 | vishva | vishva@gmail.com    | 9952364875 |
|         2 | Parii  | pari@gmail.com      | 7845963254 |
|         3 | Sakshi | chaudhari@gmail.com | 425935845  |
+-----------+--------+--------------------+------------+
3 rows in set (0.00 sec)

mysql> insert into Borrow_Records( record_id, member_id, book_id,borrow_date,return_date) values (1, 1, 1, '2026-10-01', '2026-10-10'), (2, 2, 2, '2026-10-0
2', '2026-10-12'),(3, 3, 1, '2026-10-03', '2026-10-28'),(4, 1, 5, '2026-10-05', NULL);
Query OK, 4 rows affected (0.08 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from Borrow_Records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|         1 |         1 |       1 | 2026-10-01  | 2026-10-10  |
|         2 |         2 |       2 | 2026-10-02  | 2026-10-12  |
|         3 |         3 |       1 | 2026-10-03  | 2026-10-28  |
|         4 |         1 |       5 | 2026-10-05  | NULL        |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)

mysql> select * from Books where published_year > 2015;
+---------+-----------------+-----------+----------------+-----------------+-----------+
| book_id | title           | author_id | published_year | available_copies | genre     |
+---------+-----------------+-----------+----------------+-----------------+-----------+
|       1 | Database Systems |         1 |           2018 |               5 | Technology |
|       4 | 1984             |         3 |           2019 |               6 | Dystopian  |
|       5 | SQL Fundamentals |         1 |           2022 |              10 | Technology |
+---------+-----------------+-----------+----------------+-----------------+-----------+
3 rows in set (0.07 sec)

mysql> create table Orders( order_id int primary key, member_id int, foreignkey (member_id) references Members(member_id));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(
```

```
mysql> create table Orders  (
    ->     order_id INT PRIMARY KEY,
    ->     member_id INT,
    ->     FOREIGN KEY (member_id) REFERENCES Members(member_id)
    -> );
Query OK, 0 rows affected (0.14 sec)

mysql> desc Orders;
+-----------+------+------+-----+---------+-------+
| Field     | Type | Null | Key | Default | Extra |
+-----------+------+------+-----+---------+-------+
| order_id  | int  | NO   | PRI | NULL    |       |
| member_id | int  | YES  | MUL | NULL    |       |
+-----------+------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> select distinct m.members_id, m.name, m.email from Members as m
    -> join Borrow_Records br on m.member_id = br.memner_id
    -> join Books b on br.book_id = b.book_id where b.title = 'Database Systems';
ERROR 1054 (42S22): Unknown column 'm.members_id' in 'field list'
mysql> select distinct m.member_id, m.name, m.email from Members as m
    -> join Borrow_Records br on m.member_id = br.member_id
    -> join Books b on br.book_id = b.book_id where b.title = 'Database Systems';
+-----------+--------+--------------------+
| member_id | name   | email              |
+-----------+--------+--------------------+
|         1 | vishva | vishva@gmail.com   |
|         3 | Sakshi | chaudhari@gmail.com |
+-----------+--------+--------------------+
2 rows in set (0.07 sec)

mysql> update Books set available_copies = available_copies -1
    -> where book_id =1 and available_copies>0;
Query OK, 1 row affected (0.08 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> delete from Members where member_id = 3;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`librarymanagement`.`borrow_records`, CONSTRAINT `borrow_records_i
bfk_1` FOREIGN KEY (`member_id`) REFERENCES `members` (`member_id`))
 ysql> update Books set title = 'Advanced Database Systems' where bookid =1;
ERROR 1054 (42S22): Unknown column 'bookid' in 'where clause'
mysql>
mysql>
mysql>
mysql>
mysql>
mysql> update Books set title = 'Advanced Database Systems' where book_id =1;
Query OK, 1 row affected (0.07 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select* from books;
+---------+---------------------------+-----------+----------------+-----------------+-------------------+
| book_id | title                     | author_id | published_year | available_copies | genre             |
+---------+---------------------------+-----------+----------------+-----------------+-------------------+
|       1 | Advanced Database Systems |         1 |           2018 |               4 | Technology        |
|       2 | Harry Potter              |         2 |           1997 |               8 | Fantasy           |
|       3 | Animal Farm               |         3 |           1945 |               4 | Political Fiction |
|       4 | 1984                      |         3 |           2019 |               6 | Dystopian         |
|       5 | SQL Fundamentals          |         1 |           2022 |              10 | Technology        |
+---------+---------------------------+-----------+----------------+-----------------+-------------------+
5 rows in set (0.00 sec)

mysql> select b.book_id, b.title, a.name as Author_name, b.published_year from books b
    -> join Authors a on b.author_id = a.author_id;
+---------+---------------------------+--------------+----------------+
| book_id | title                     | Author_name  | published_year |
+---------+---------------------------+--------------+----------------+
|       1 | Advanced Database Systems | R. K. Narayan |           2018 |
|       5 | SQL Fundamentals          | R. K. Narayan |           2022 |
|       2 | Harry Potter              | J. K. Rowling |           1997 |
|       3 | Animal Farm               | George Orwell |           1945 |
|       4 | 1984                      | George Orwell |           2019 |
+---------+---------------------------+--------------+----------------+
5 rows in set (0.00 sec)

mysql> delete from books where published_year <2000;
```

```
mysql> SELECT b.*
    -> FROM Books b
    -> LEFT JOIN Borrow_Records br
    ->     ON b.book_id = br.book_id
    -> WHERE br.book_id IS NULL;
+---------+-------------+-----------+----------------+-----------------+-------------------+
| book_id | title       | author_id | published_year | available_copies | genre             |
+---------+-------------+-----------+----------------+-----------------+-------------------+
|       3 | Animal Farm |         3 |           1945 |               4 | Political Fiction |
|       4 | 1984        |         3 |           2019 |               6 | Dystopian         |
+---------+-------------+-----------+----------------+-----------------+-------------------+
2 rows in set (0.03 sec)

mysql>
```